

Database Fundamental

Kelompok 2

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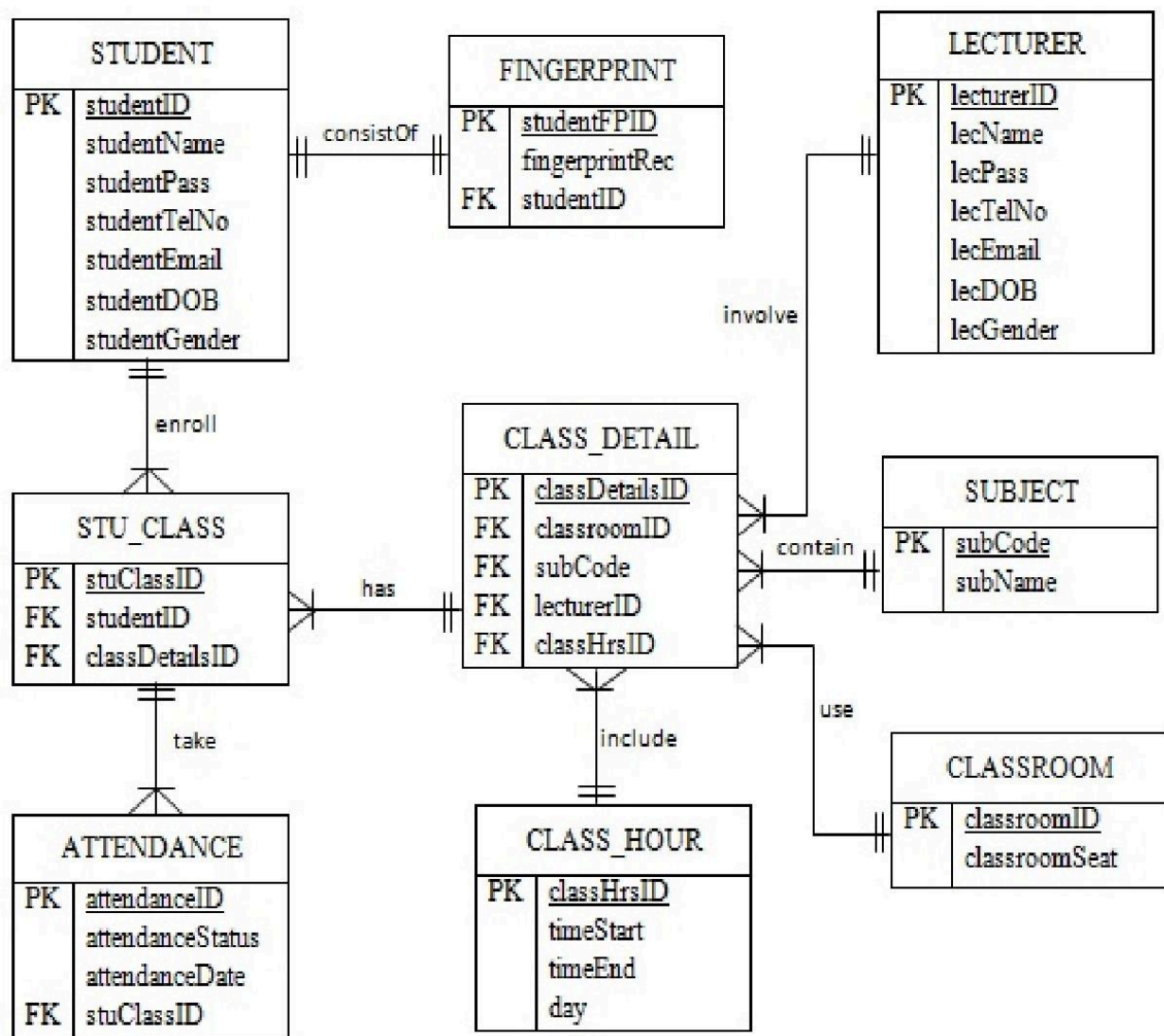
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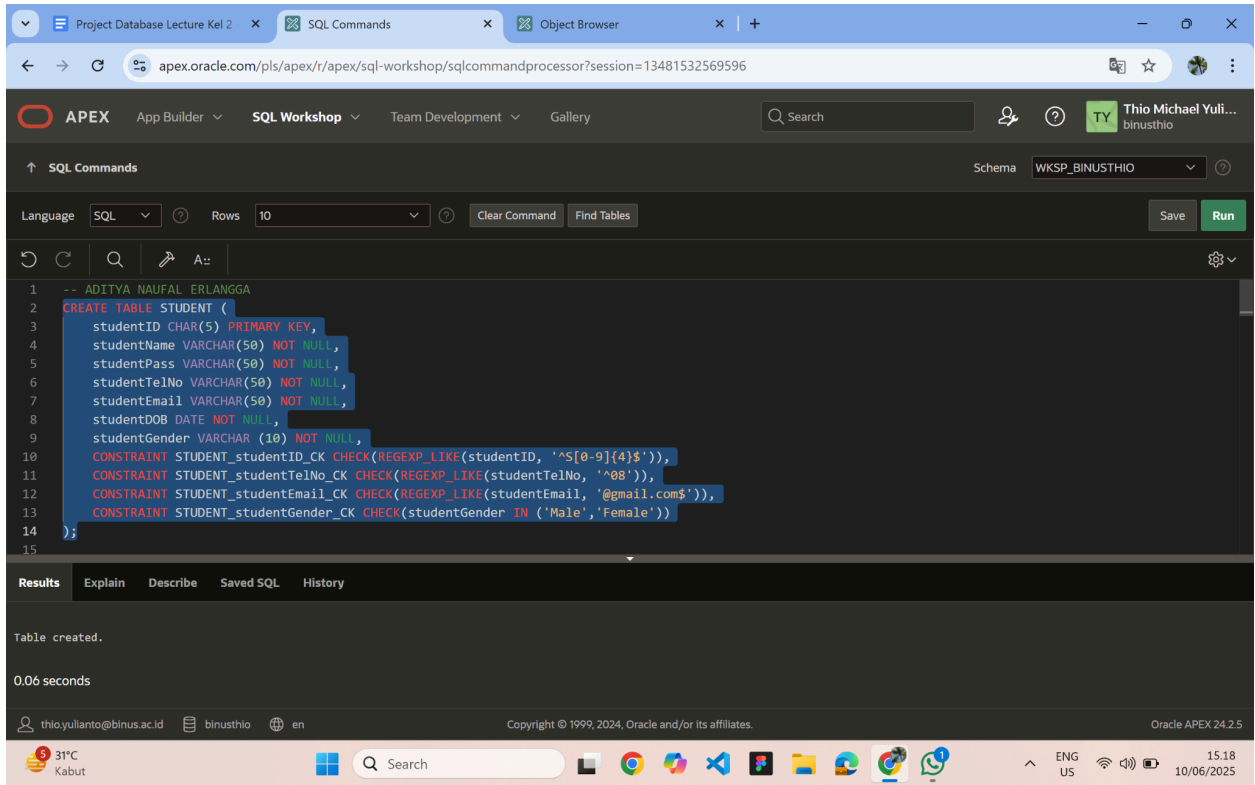
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University Academic & Attendance Management System Database Design

Gambar ERD Kelompok



CREATE TABLE STUDENT (ADITYA NAUFAL ERLANGGA):



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli...' with the schema 'WKSP_BINUSTHIO'. The 'SQL Commands' tab is active, showing a SQL command to create a table named 'STUDENT'. The command is as follows:

```
1  -- ADITYA NAUFAL ERLANGGA
2  CREATE TABLE STUDENT (
3      studentID CHAR(5) PRIMARY KEY,
4      studentName VARCHAR(50) NOT NULL,
5      studentPass VARCHAR(50) NOT NULL,
6      studentTelNo VARCHAR(50) NOT NULL,
7      studentEmail VARCHAR(50) NOT NULL,
8      studentDOB DATE NOT NULL,
9      studentGender VARCHAR(10) NOT NULL,
10     CONSTRAINT STUDENT_studentID_CK CHECK(REGEXP_LIKE(studentID, '^S[0-9]{4}$')),
11     CONSTRAINT STUDENT_studentTelNo_CK CHECK(REGEXP_LIKE(studentTelNo, '^08')),
12     CONSTRAINT STUDENT_studentEmail_CK CHECK(REGEXP_LIKE(studentEmail, '@gmail.com$')),
13     CONSTRAINT STUDENT_studentGender_CK CHECK(studentGender IN ('Male','Female'))
14 );
15
```

The 'Results' tab shows the message 'Table created.' and the execution time '0.06 seconds'.

QUERY:

-- ADITYA NAUFAL ERLANGGA

```
CREATE TABLE STUDENT (
    studentID CHAR(5) PRIMARY KEY,
    studentName VARCHAR(50) NOT NULL,
    studentPass VARCHAR(50) NOT NULL,
    studentTelNo VARCHAR(50) NOT NULL,
    studentEmail VARCHAR(50) NOT NULL,
    studentDOB DATE NOT NULL,
    studentGender VARCHAR(10) NOT NULL,
    CONSTRAINT STUDENT_studentID_CK CHECK(REGEXP_LIKE(studentID,
'^S[0-9]{4}$')),
    CONSTRAINT STUDENT_studentTelNo_CK CHECK(REGEXP_LIKE(studentTelNo,
'^08')),
    CONSTRAINT STUDENT_studentEmail_CK CHECK(REGEXP_LIKE(studentEmail,
'@gmail.com$')),
    CONSTRAINT STUDENT_studentGender_CK CHECK(studentGender IN ('Male','Female'))
);
```

CREATE TABLE FINGERPRINT (ADITYA NAUFAL ERLANGGA):

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli...' with the schema 'WKSP_BINUSTHIO'. The 'SQL Commands' tab is active, showing a SQL command editor with the following code:

```
-- ADITYA NAUFAL ERLANGGA
CREATE TABLE FINGERPRINT(
  studentFPID CHAR(5) PRIMARY KEY,
  fingerprintRec BLOB NOT NULL,
  studentID CHAR(5) NOT NULL,
  CONSTRAINT FINGERPRINT_studentFPID_CK CHECK(REGEXP_LIKE(studentFPID, '^FP[0-9]{3}$')),
  CONSTRAINT FINGERPRINT_studentFPID_FK FOREIGN KEY (studentID) REFERENCES STUDENT (studentID)
);
```

The 'Run' button is highlighted in green. Below the editor, the 'Results' tab is selected, displaying the message 'Table created.' and the execution time '0.07 seconds'.

QUERY:

-- ADITYA NAUFAL ERLANGGA

```
CREATE TABLE FINGERPRINT(
  studentFPID CHAR(5) PRIMARY KEY,
  fingerprintRec BLOB NOT NULL,
  studentID CHAR(5) NOT NULL,
  CONSTRAINT FINGERPRINT_studentFPID_CK CHECK(REGEXP_LIKE(studentFPID,
'^FP[0-9]{3}$')),
  CONSTRAINT FINGERPRINT_studentFPID_FK FOREIGN KEY (studentID)
REFERENCES STUDENT (studentID)
);
```

CREATE TABLE LECTURER (ARYA RAKA PRATAMA):

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, displaying a SQL script for creating a table named 'LECTURER'. The script includes columns for lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, and lecGender, along with several CHECK constraints. The 'Run' button is highlighted in green. Below the script, the 'Results' tab shows the message 'Table created.' and a duration of '0.05 seconds'. The bottom status bar indicates the user 'thio.yulianto@binus.ac.id' and the schema 'binusthio'.

```
-- ARYA RAKA PRATAMA
CREATE TABLE LECTURER(
  lecturerID CHAR(5) PRIMARY KEY,
  lecName VARCHAR(50) NOT NULL,
  lecPass VARCHAR(50) NOT NULL,
  lecTelNo VARCHAR(50) NOT NULL,
  lecEmail VARCHAR(50) NOT NULL,
  lecDOB DATE NOT NULL,
  lecGender VARCHAR (10) NOT NULL,
  CONSTRAINT LECTURER_lecturerID_CK CHECK(REGEXP_LIKE(lecturerID, '^L[0-9]{4}$')),
  CONSTRAINT LECTURER_lectelNo_CK CHECK(REGEXP_LIKE(lecTelNo, '^08')),
  CONSTRAINT LECTURER_lecEmail_CK CHECK(REGEXP_LIKE(lecEmail, '@gmail.com$')),
  CONSTRAINT LECTURER_lecGender_CK CHECK(lecGender IN ('Male','Female'))
);
```

Results Explain Describe Saved SQL History

Table created.

0.05 seconds

thio.yulianto@binus.ac.id binusthio en Copyright © 1999, 2024, Oracle and/or its affiliates. Oracle APEX 24.2.5

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QUERY:

-- ARYA RAKA PRATAMA

```
CREATE TABLE LECTURER(
  lecturerID CHAR(5) PRIMARY KEY,
  lecName VARCHAR(50) NOT NULL,
  lecPass VARCHAR(50) NOT NULL,
  lecTelNo VARCHAR(50) NOT NULL,
  lecEmail VARCHAR(50) NOT NULL,
  lecDOB DATE NOT NULL,
  lecGender VARCHAR (10) NOT NULL,
  CONSTRAINT LECTURER_lecturerID_CK CHECK(REGEXP_LIKE(lecturerID,
'^L[0-9]{4}$')),
  CONSTRAINT LECTURER_lectelNo_CK CHECK(REGEXP_LIKE(lecTelNo, '^08')),
  CONSTRAINT LECTURER_lecEmail_CK CHECK(REGEXP_LIKE(lecEmail,
'@gmail.com$')),
  CONSTRAINT LECTURER_lecGender_CK CHECK(lecGender IN ('Male','Female'))
);
```

CREATE TABLE SUBJECT (SAMUEL PRIMA DAMANIK):

The screenshot shows the Oracle APEX SQL Workshop interface. The browser address bar displays the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=13481532569596`. The interface includes a top navigation bar with tabs for 'Project Database Lecture Kel 2', 'SQL Commands', and 'Object Browser'. The main area is titled 'SQL Commands' and shows a schema dropdown set to 'WKSP_BINUSTHIO'. The SQL editor contains the following code:

```
36 CONSTRAINT LECTURER_lectemail_CK CHECK(REGEXP_LIKE(lectemail, @gmail.com$)),
37 CONSTRAINT LECTURER_lectGender_CK CHECK(lectGender IN ('Male','Female'))
38 );
39
40 -- SAMUEL PRIMA DAMANIK
41 CREATE TABLE SUBJECT(
42     subCode CHAR(5) PRIMARY KEY,
43     subName VARCHAR(50) NOT NULL,
44     CONSTRAINT SUBJECT_subCode_CK CHECK(REGEXP_LIKE(subCode, '^SUB[0-9]{2}$'))
45 );
46
47 -- SAMUEL PRIMA DAMANIK
48 CREATE TABLE CLASSROOM(
49     classroomID CHAR(4) PRIMARY KEY,
50     classroomSeat NUMBER(3,0) NOT NULL,
```

Below the editor, the 'Results' tab is active, displaying the message 'Table created.' and the execution time '0.06 seconds'. The bottom status bar shows the user 'thio.yulianto@binus.ac.id', the schema 'binusthio', and the Oracle APEX version '24.2.5'.

QUERY:

-- SAMUEL PRIMA DAMANIK

CREATE TABLE SUBJECT(

subCode CHAR(5) PRIMARY KEY,

subName VARCHAR(50) NOT NULL,

CONSTRAINT SUBJECT_subCode_CK CHECK(REGEXP_LIKE(subCode,
'^SUB[0-9]{2}\$'))

);

CREATE TABLE CLASSROOM (SAMUEL PRIMA DAMANIK):

The screenshot shows the Oracle APEX SQL Workshop interface. The browser address bar displays the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=13481532569596`. The interface includes a top navigation bar with tabs for 'Project Database Lecture Kel 2', 'SQL Commands', and 'Object Browser'. The main area is titled 'SQL Commands' and shows a schema dropdown set to 'WKSP_BINUSTHIO'. The 'Language' is set to 'SQL' and 'Rows' is set to '10'. The 'Run' button is highlighted in green. The SQL command being executed is:

```
42 subCode CHAR(5) PRIMARY KEY,  
43 subName VARCHAR(50) NOT NULL,  
44 CONSTRAINT SUBJECT_subCode_CK CHECK(REGEXP_LIKE(subCode, '^SUB[0-9]{2}$'))  
45 );  
46  
47 -- SAMUEL PRIMA DAMANIK  
48 CREATE TABLE CLASSROOM(  
49 classroomID CHAR(4) PRIMARY KEY,  
50 classroomSeat NUMBER(3,0) NOT NULL,  
51 CONSTRAINT CLASSROOM_classroomID_CK CHECK(REGEXP_LIKE(classroomID, '^C[0-9]{3}$'))  
52 );  
53  
54 -- ANDHIKA HAFIZH ALBANA  
55 CREATE TABLE CLASS_HOUR(  
56 classHrsID CHAR(5) PRIMARY KEY,
```

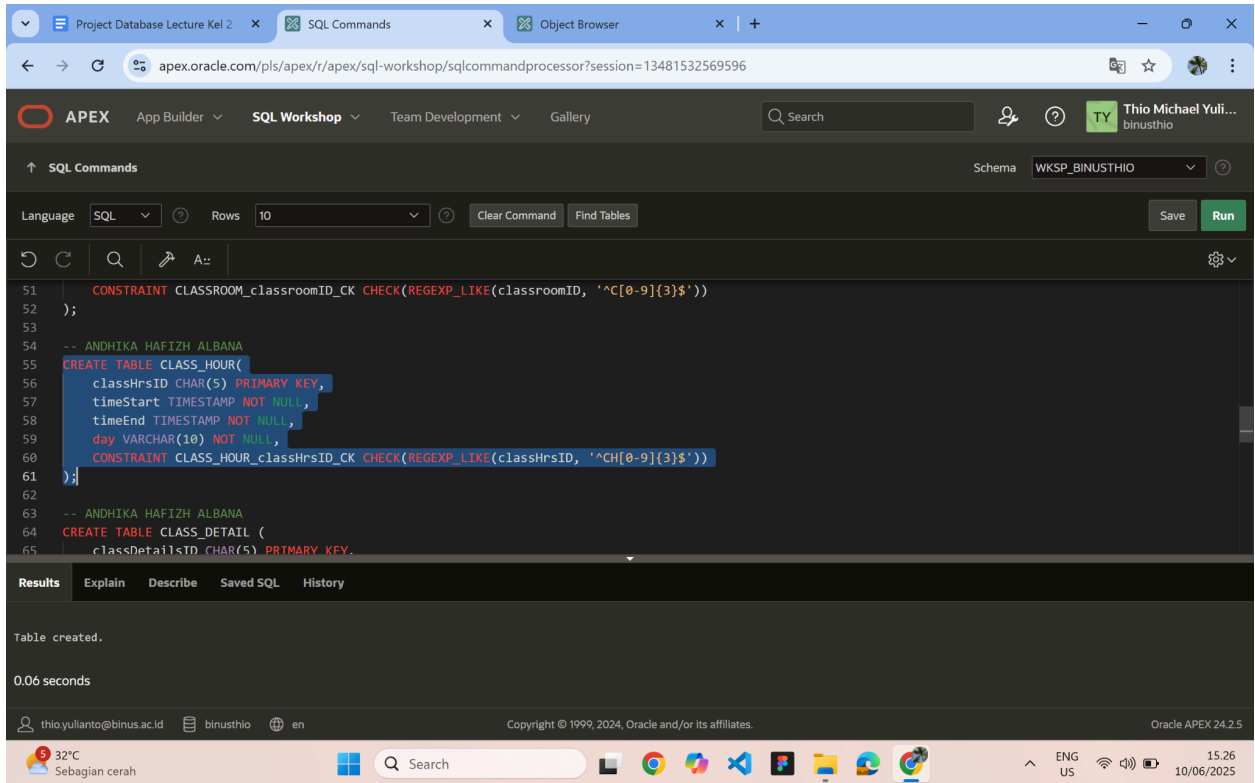
The 'Results' tab is active, showing the message 'Table created.' and a duration of '0.05 seconds'. The bottom status bar indicates the user is 'thio.yulianto@binus.ac.id', the schema is 'binusthio', and the language is 'en'. The Oracle APEX version is 24.2.5. The system clock shows 15:24 on 10/06/2025.

QUERY:

-- SAMUEL PRIMA DAMANIK

```
CREATE TABLE CLASSROOM(  
    classroomID CHAR(4) PRIMARY KEY,  
    classroomSeat NUMBER(3,0) NOT NULL,  
    CONSTRAINT CLASSROOM_classroomID_CK CHECK(REGEXP_LIKE(classroomID,  
'^C[0-9]{3}$'))  
);
```

CREATE TABLE CLASS_HOUR (ANDHIKA HAFIZH ALBANA):



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, displaying a SQL command editor. The command is as follows:

```
51 CONSTRAINT CLASSROOM_classroomID_CK CHECK(REGEXP_LIKE(classroomID, '^C[0-9]{3}$'))
52 );
53
54 -- ANDHIKA HAFIZH ALBANA
55 CREATE TABLE CLASS_HOUR(
56   classHrsID CHAR(5) PRIMARY KEY,
57   timeStart TIMESTAMP NOT NULL,
58   timeEnd TIMESTAMP NOT NULL,
59   day VARCHAR(10) NOT NULL,
60   CONSTRAINT CLASS_HOUR_classHrsID_CK CHECK(REGEXP_LIKE(classHrsID, '^CH[0-9]{3}$'))
61 );
62
63 -- ANDHIKA HAFIZH ALBANA
64 CREATE TABLE CLASS_DETAIL (
65   classDetailsID CHAR(5) PRIMARY KEY,
```

The 'Run' button is highlighted in green. Below the editor, the 'Results' tab is active, showing the message 'Table created.' and a duration of '0.06 seconds'. The bottom status bar indicates the user 'thio.yulianto@binus.ac.id', the schema 'binusthio', and the Oracle APEX version '24.2.5'.

QUERY:

```
-- ANDHIKA HAFIZH ALBANA
CREATE TABLE CLASS_HOUR(
  classHrsID CHAR(5) PRIMARY KEY,
  timeStart TIMESTAMP NOT NULL,
  timeEnd TIMESTAMP NOT NULL,
  day VARCHAR(10) NOT NULL,
  CONSTRAINT CLASS_HOUR_classHrsID_CK CHECK(REGEXP_LIKE(classHrsID,
'^CH[0-9]{3}$'))
);
```

CREATE TABLE CLASS_DETAIL (ANDHIKA HAFIZH ALBANA):

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, displaying a SQL command to create a table named CLASS_DETAIL. The command is as follows:

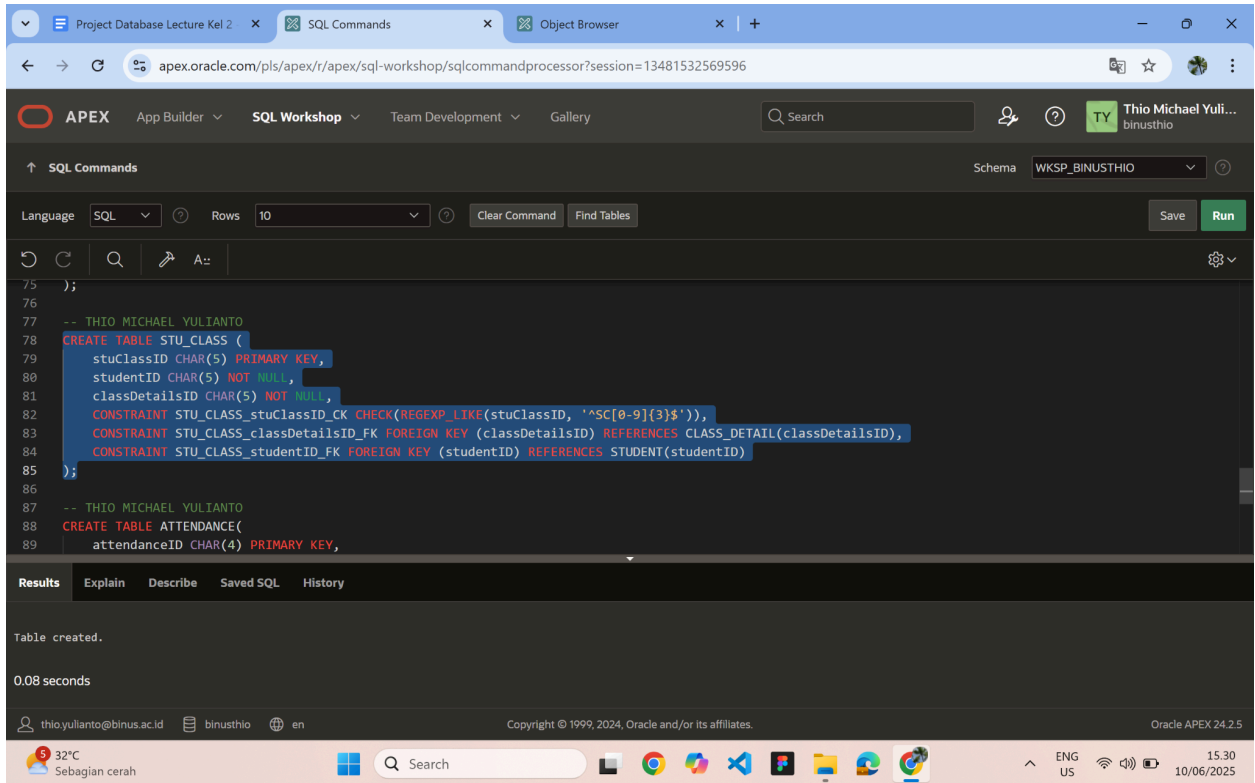
```
-- ANDHIKA HAFIZH ALBANA
CREATE TABLE CLASS_DETAIL (
  classDetailsID CHAR(5) PRIMARY KEY,
  classroomID CHAR(4) NOT NULL,
  subCode CHAR(5) NOT NULL,
  lecturerID CHAR(5) NOT NULL,
  classHrsID CHAR(5) NOT NULL,
  CONSTRAINT CLASS_DETAIL_classDetailsID_CK CHECK(REGEXP_LIKE(classDetailsID, '^CD[0-9]{3}$')),
  CONSTRAINT CLASS_DETAIL_classroomID_FK FOREIGN KEY (classroomID) REFERENCES CLASSROOM (classroomID),
  CONSTRAINT CLASS_DETAIL_subCode_FK FOREIGN KEY (subCode) REFERENCES SUBJECT (subCode),
  CONSTRAINT CLASS_DETAIL_lecturerID_FK FOREIGN KEY (lecturerID) REFERENCES LECTURER (lecturerID),
  CONSTRAINT CLASS_DETAIL_classHrsID_FK FOREIGN KEY (classHrsID) REFERENCES CLASS_HOUR (classHrsID)
);
```

The 'Results' tab at the bottom shows the message 'Table created.' and the execution time '0.07 seconds'. The user's name 'Thio Michael Yuli...' and the schema 'WKSP_BINUSTHIO' are visible in the top right corner.

QUERY:

```
-- ANDHIKA HAFIZH ALBANA
CREATE TABLE CLASS_DETAIL (
  classDetailsID CHAR(5) PRIMARY KEY,
  classroomID CHAR(4) NOT NULL,
  subCode CHAR(5) NOT NULL,
  lecturerID CHAR(5) NOT NULL,
  classHrsID CHAR(5) NOT NULL,
  CONSTRAINT CLASS_DETAIL_classDetailsID_CK
CHECK(REGEXP_LIKE(classDetailsID, '^CD[0-9]{3}$')),
  CONSTRAINT CLASS_DETAIL_classroomID_FK FOREIGN KEY (classroomID)
REFERENCES CLASSROOM (classroomID),
  CONSTRAINT CLASS_DETAIL_subCode_FK FOREIGN KEY (subCode) REFERENCES
SUBJECT (subCode),
  CONSTRAINT CLASS_DETAIL_lecturerID_FK FOREIGN KEY (lecturerID)
REFERENCES LECTURER (lecturerID),
  CONSTRAINT CLASS_DETAIL_classHrsID_FK FOREIGN KEY (classHrsID)
REFERENCES CLASS_HOUR (classHrsID)
);
```

CREATE TABLE STU_CLASS (THIO MICHAEL YULIANTO):



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yulianto' (binusthio). The 'SQL Commands' tab is active, showing a SQL command editor with the following code:

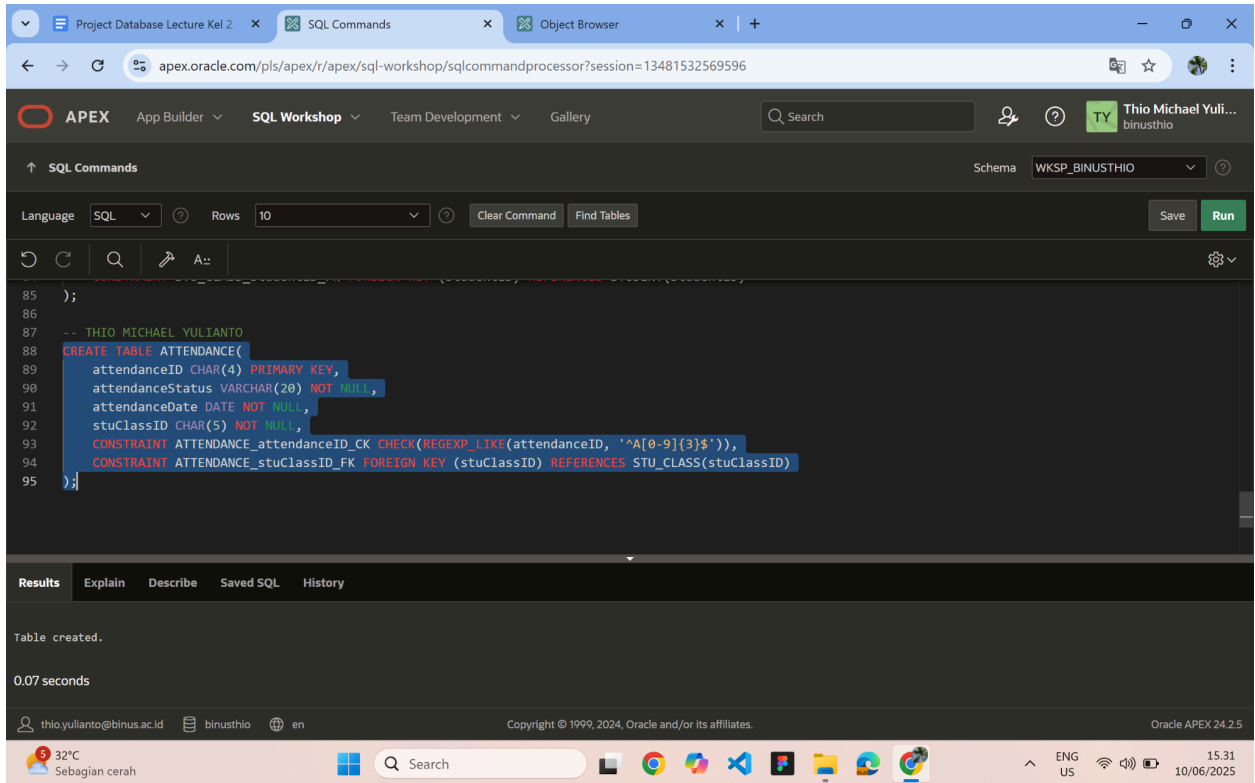
```
-- THIO MICHAEL YULIANTO
CREATE TABLE STU_CLASS (
  stuClassID CHAR(5) PRIMARY KEY,
  studentID CHAR(5) NOT NULL,
  classDetailsID CHAR(5) NOT NULL,
  CONSTRAINT STU_CLASS_stuClassID_CK CHECK(REGEXP_LIKE(stuClassID, '^SC[0-9]{3}$')),
  CONSTRAINT STU_CLASS_classDetailsID_FK FOREIGN KEY (classDetailsID) REFERENCES CLASS_DETAIL(classDetailsID),
  CONSTRAINT STU_CLASS_studentID_FK FOREIGN KEY (studentID) REFERENCES STUDENT(studentID)
);
```

The 'Run' button is highlighted in green. Below the editor, the 'Results' tab is active, displaying the message 'Table created.' and the execution time '0.08 seconds'. The bottom status bar shows the user's email 'thio.yulianto@binus.ac.id', the schema 'binusthio', and the Oracle APEX version '24.2.5'.

QUERY:

```
-- THIO MICHAEL YULIANTO
CREATE TABLE STU_CLASS (
  stuClassID CHAR(5) PRIMARY KEY,
  studentID CHAR(5) NOT NULL,
  classDetailsID CHAR(5) NOT NULL,
  CONSTRAINT STU_CLASS_stuClassID_CK CHECK(REGEXP_LIKE(stuClassID,
'^SC[0-9]{3}$')),
  CONSTRAINT STU_CLASS_classDetailsID_FK FOREIGN KEY (classDetailsID)
REFERENCES CLASS_DETAIL(classDetailsID),
  CONSTRAINT STU_CLASS_studentID_FK FOREIGN KEY (studentID) REFERENCES
STUDENT(studentID)
);
```

CREATE TABLE ATTENDANCE (THIO MICHAEL YULIANTO):



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yulianto' (binusthio). The 'SQL Commands' tab is active, showing a schema of 'WKSP_BINUSTHIO'. The SQL editor contains the following code:

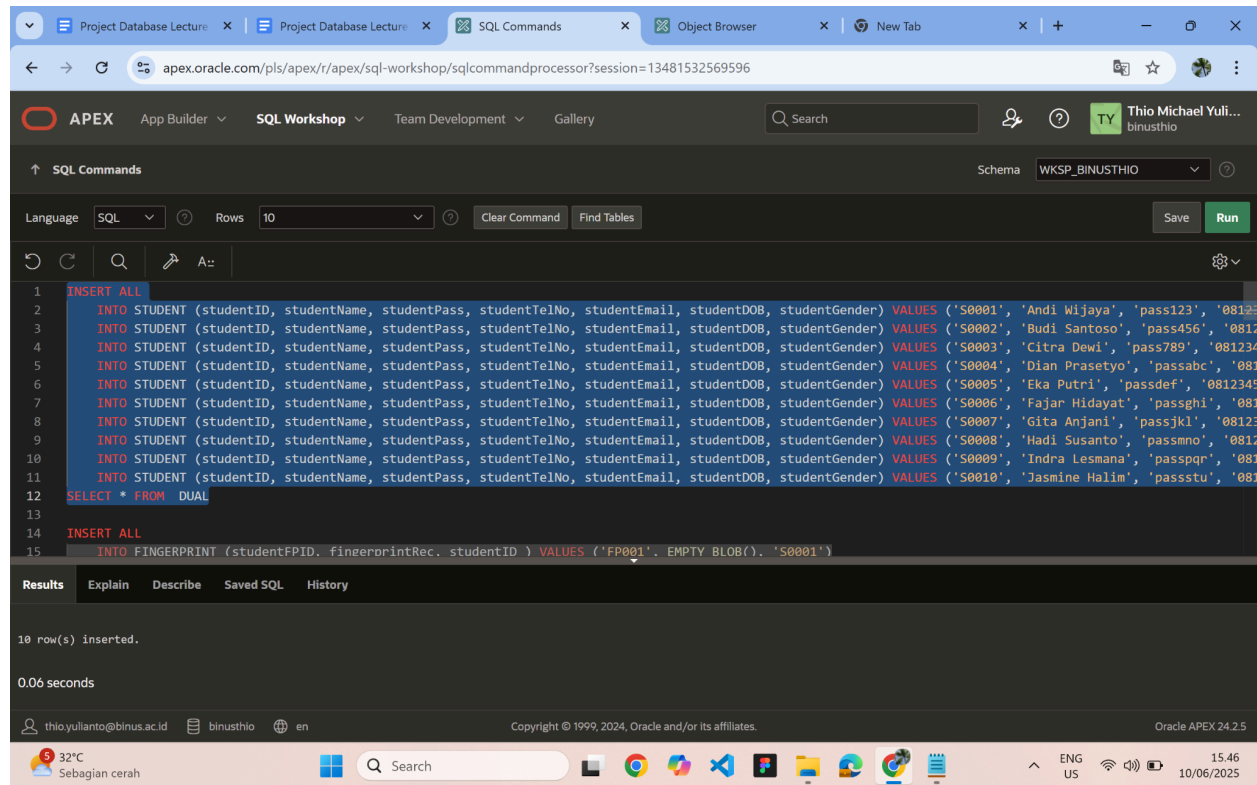
```
85 );
86
87 -- THIO MICHAEL YULIANTO
88 CREATE TABLE ATTENDANCE(
89     attendanceID CHAR(4) PRIMARY KEY,
90     attendanceStatus VARCHAR(20) NOT NULL,
91     attendanceDate DATE NOT NULL,
92     stuClassID CHAR(5) NOT NULL,
93     CONSTRAINT ATTENDANCE_attendanceID_CK CHECK(REGEXP_LIKE(attendanceID, '^A[0-9]{3}$')),
94     CONSTRAINT ATTENDANCE_stuClassID_FK FOREIGN KEY (stuClassID) REFERENCES STU_CLASS(stuClassID)
95 );
```

The 'Results' tab is selected, displaying the message 'Table created.' and the execution time '0.07 seconds'. The bottom status bar shows the user 'thio.yulianto@binus.ac.id', the schema 'binusthio', and the Oracle APEX version '24.2.5'.

QUERY:

```
-- THIO MICHAEL YULIANTO
CREATE TABLE ATTENDANCE(
    attendanceID CHAR(4) PRIMARY KEY,
    attendanceStatus VARCHAR(20) NOT NULL,
    attendanceDate DATE NOT NULL,
    stuClassID CHAR(5) NOT NULL,
    CONSTRAINT ATTENDANCE_attendanceID_CK CHECK(REGEXP_LIKE(attendanceID,
'^A[0-9]{3}$')),
    CONSTRAINT ATTENDANCE_stuClassID_FK FOREIGN KEY (stuClassID)
REFERENCES STU_CLASS(stuClassID)
);
```

INSERT TABLE STUDENT (ADITYA NAUFAL ERLANGGA):



The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command is as follows:

```
1 INSERT ALL
2 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0001', 'Andi Wijaya', 'pass123', '081234567890',
3 'andi@gmail.com', TO_DATE('2000-06-15', 'YYYY-MM-DD'), 'Male')
4 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0002', 'Budi Santoso', 'pass456', '081234567891',
5 'budi@gmail.com', TO_DATE('2001-03-10', 'YYYY-MM-DD'), 'Male')
6 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0003', 'Citra Dewi', 'pass789', '081234567892',
7 'citra@gmail.com', TO_DATE('2002-08-20', 'YYYY-MM-DD'), 'Female')
8 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0004', 'Dian Prasetyo', 'passabc', '081234567893',
9 'dian@gmail.com', TO_DATE('2000-12-05', 'YYYY-MM-DD'), 'Male')
10 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0005', 'Eka Putri', 'passdef', '081234567894',
11 'eka@gmail.com', TO_DATE('1999-07-25', 'YYYY-MM-DD'), 'Female')
12 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0006', 'Fajar Hidayat', 'passghi', '081234567895',
13 'fajar@gmail.com', TO_DATE('2003-11-30', 'YYYY-MM-DD'), 'Male')
14 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0007', 'Gita Anjani', 'passjkl', '081234567896',
15 'gita@gmail.com', TO_DATE('2004-04-15', 'YYYY-MM-DD'), 'Female')
16 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0008', 'Hadi Susanto', 'passmno', '081234567897',
17 'hadi@gmail.com', TO_DATE('2005-09-01', 'YYYY-MM-DD'), 'Male')
18 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0009', 'Indra Lesmana', 'passpqr', '081234567898',
19 'indra@gmail.com', TO_DATE('2006-02-18', 'YYYY-MM-DD'), 'Male')
20 INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0010', 'Jasmine Halim', 'passstu', '081234567899',
21 'jasmine@gmail.com', TO_DATE('2007-05-22', 'YYYY-MM-DD'), 'Female')
22 SELECT * FROM DUAL
```

The results section shows:

10 row(s) inserted.

0.06 seconds

QUERY:

INSERT ALL

INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0001', 'Andi Wijaya', 'pass123', '081234567890', 'andi@gmail.com', TO_DATE('2000-06-15', 'YYYY-MM-DD'), 'Male')

INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0002', 'Budi Santoso', 'pass456', '081234567891', 'budi@gmail.com', TO_DATE('2001-03-10', 'YYYY-MM-DD'), 'Male')

INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0003', 'Citra Dewi', 'pass789', '081234567892', 'citra@gmail.com', TO_DATE('2002-08-20', 'YYYY-MM-DD'), 'Female')

INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0004', 'Dian Prasetyo', 'passabc', '081234567893', 'dian@gmail.com', TO_DATE('2000-12-05', 'YYYY-MM-DD'), 'Male')

INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0005', 'Eka Putri', 'passdef', '081234567894', 'eka@gmail.com', TO_DATE('1999-07-25', 'YYYY-MM-DD'), 'Female')

INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail, studentDOB, studentGender) VALUES ('S0006', 'Fajar Hidayat', 'passghi', '081234567895', 'fajar@gmail.com', TO_DATE('2003-11-30', 'YYYY-MM-DD'), 'Male')

```
INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail,
studentDOB, studentGender) VALUES ('S0007', 'Gita Anjani', 'passjkl', '081234567896',
'gita@gmail.com', TO_DATE('2001-05-18', 'YYYY-MM-DD'), 'Female')
```

```
INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail,
studentDOB, studentGender) VALUES ('S0008', 'Hadi Susanto', 'passmno', '081234567897',
'hadi@gmail.com', TO_DATE('2000-09-22', 'YYYY-MM-DD'), 'Male')
```

```
INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail,
studentDOB, studentGender) VALUES ('S0009', 'Indra Lesmana', 'passpqr', '081234567898',
'indra@gmail.com', TO_DATE('1998-04-12', 'YYYY-MM-DD'), 'Male')
```

```
INTO STUDENT (studentID, studentName, studentPass, studentTelNo, studentEmail,
studentDOB, studentGender) VALUES ('S0010', 'Jasmine Halim', 'passstu', '081234567899',
'jasmine@gmail.com', TO_DATE('2002-02-08', 'YYYY-MM-DD'), 'Female')
```

```
SELECT * FROM DUAL
```

INSERT TABLE FINGERPRINT (ADITYA NAUFAL ERLANGGA):

The screenshot shows the Oracle APEX SQL Workshop interface. The command window contains the following SQL statements:

```
INSERT ALL
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP001', EMPTY_BLOB(), 'S0001')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP002', EMPTY_BLOB(), 'S0002')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP003', EMPTY_BLOB(), 'S0003')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP004', EMPTY_BLOB(), 'S0004')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP005', EMPTY_BLOB(), 'S0005')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP006', EMPTY_BLOB(), 'S0006')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP007', EMPTY_BLOB(), 'S0007')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP008', EMPTY_BLOB(), 'S0008')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP009', EMPTY_BLOB(), 'S0009')
  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP010', EMPTY_BLOB(), 'S0010')
SELECT * FROM DUAL
```

The results pane shows:

```
10 row(s) inserted.
0.06 seconds
```

QUERY:

INSERT ALL

```
INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP001',
EMPTY_BLOB(), 'S0001')
```

```
INTO FINGERPRINT (studentFPID, fingerprintRec, studentID) VALUES ('FP002',
EMPTY_BLOB(), 'S0002')
```

```

    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP003',
EMPTY_BLOB(), 'S0003')
    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP004',
EMPTY_BLOB(), 'S0004')
    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP005',
EMPTY_BLOB(), 'S0005')
    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP006',
EMPTY_BLOB(), 'S0006')
    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP007',
EMPTY_BLOB(), 'S0007')
    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP008',
EMPTY_BLOB(), 'S0008')
    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP009',
EMPTY_BLOB(), 'S0009')
    INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP010',
EMPTY_BLOB(), 'S0010')
SELECT * FROM DUAL

```

INSERT TABLE LECTURER (ARYA RAKA PRATAMA):

The screenshot shows the Oracle APEX SQL Workshop interface. The 'SQL Commands' tab is active, displaying a series of SQL statements. The first part of the script inserts data into the FINGERPRINT table for student IDs S0003 through S0010. The second part inserts data into the LECTURER table for five lecturers. The third part inserts data into the SUBJECT table. The 'Run' button has been clicked, and the 'Results' tab shows '5 row(s) inserted.' and '0.05 seconds'.

```

23  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP009', EMPTY_BLOB(), 'S0009')
24  INTO FINGERPRINT (studentFPID, fingerprintRec, studentID ) VALUES ('FP010', EMPTY_BLOB(), 'S0010')
25  SELECT * FROM DUAL
26
27  INSERT ALL
28  INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender) VALUES ('L0001', 'Dr. Ahmad Fauzi', 'lec123', '08121111111', 'ahmad@gma
29  INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender) VALUES ('L0002', 'Prof. Susan Dewi', 'lec456', '08121111112', 'susan@gn
30  INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender) VALUES ('L0003', 'Dr. Bambang Setiawan', 'lec789', '08121111113', 'bamb
31  INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender) VALUES ('L0004', 'Dr. Kartika Sari', 'lecabc', '08121111114', 'kartikag
32  INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender) VALUES ('L0005', 'Dr. Rudi Hartono', 'lecdef', '08121111115', 'rudi@gma
33  SELECT * FROM DUAL
34
35  INSERT ALL
36  INTO SUBJECT (subCode, subName) VALUES ('SUB01', 'Database Systems')

```

Results: 5 row(s) inserted.
0.05 seconds

QUERY:

INSERT ALL

```
INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender)
VALUES ('L0001', 'Dr. Ahmad Fauzi', 'lec123', '08121111111', 'ahmad@gmail.com',
TO_DATE('1980-05-12', 'YYYY-MM-DD'), 'Male')
```

```
INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender)
VALUES ('L0002', 'Prof. Susan Dewi', 'lec456', '08121111112', 'susan@gmail.com',
TO_DATE('1975-09-18', 'YYYY-MM-DD'), 'Female')
```

```
INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender)
VALUES ('L0003', 'Dr. Bambang Setiawan', 'lec789', '08121111113', 'bambang@gmail.com',
TO_DATE('1982-03-20', 'YYYY-MM-DD'), 'Male')
```

```
INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender)
VALUES ('L0004', 'Dr. Kartika Sari', 'lecabc', '08121111114', 'kartika@gmail.com',
TO_DATE('1978-07-30', 'YYYY-MM-DD'), 'Female')
```

```
INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender)
VALUES ('L0005', 'Dr. Rudi Hartono', 'lecdef', '08121111115', 'rudi@gmail.com',
TO_DATE('1985-01-10', 'YYYY-MM-DD'), 'Male')
```

```
SELECT * FROM DUAL
```

INSERT TABLE SUBJECT (SAMUEL PRIMA DAMANIK):

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, displaying a list of SQL commands. The commands are as follows:

```
31 INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender) VALUES ('L0004', 'Dr. Kartika Sari', 'lecabc', '08121111114', 'kartika@gmail.com',
32 INTO LECTURER (lecturerID, lecName, lecPass, lecTelNo, lecEmail, lecDOB, lecGender) VALUES ('L0005', 'Dr. Rudi Hartono', 'lecdef', '08121111115', 'rudi@gmail.com',
33 SELECT * FROM DUAL
34
35 INSERT ALL
36 INTO SUBJECT (subCode, subName) VALUES ('SUB01', 'Database Systems')
37 INTO SUBJECT (subCode, subName) VALUES ('SUB02', 'Artificial Intelligence')
38 INTO SUBJECT (subCode, subName) VALUES ('SUB03', 'Software Engineering')
39 INTO SUBJECT (subCode, subName) VALUES ('SUB04', 'Computer Networks')
40 INTO SUBJECT (subCode, subName) VALUES ('SUB05', 'Operating Systems')
41 SELECT * FROM DUAL
42
43 INSERT ALL
44 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C001', 50)
45 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C002', 40)
```

The 'Results' tab shows the execution results: '5 row(s) inserted.' and '0.03 seconds'. The bottom status bar indicates the user is 'thio.yulianto@binus.ac.id' and the session is 'binusthio'.

QUERY:

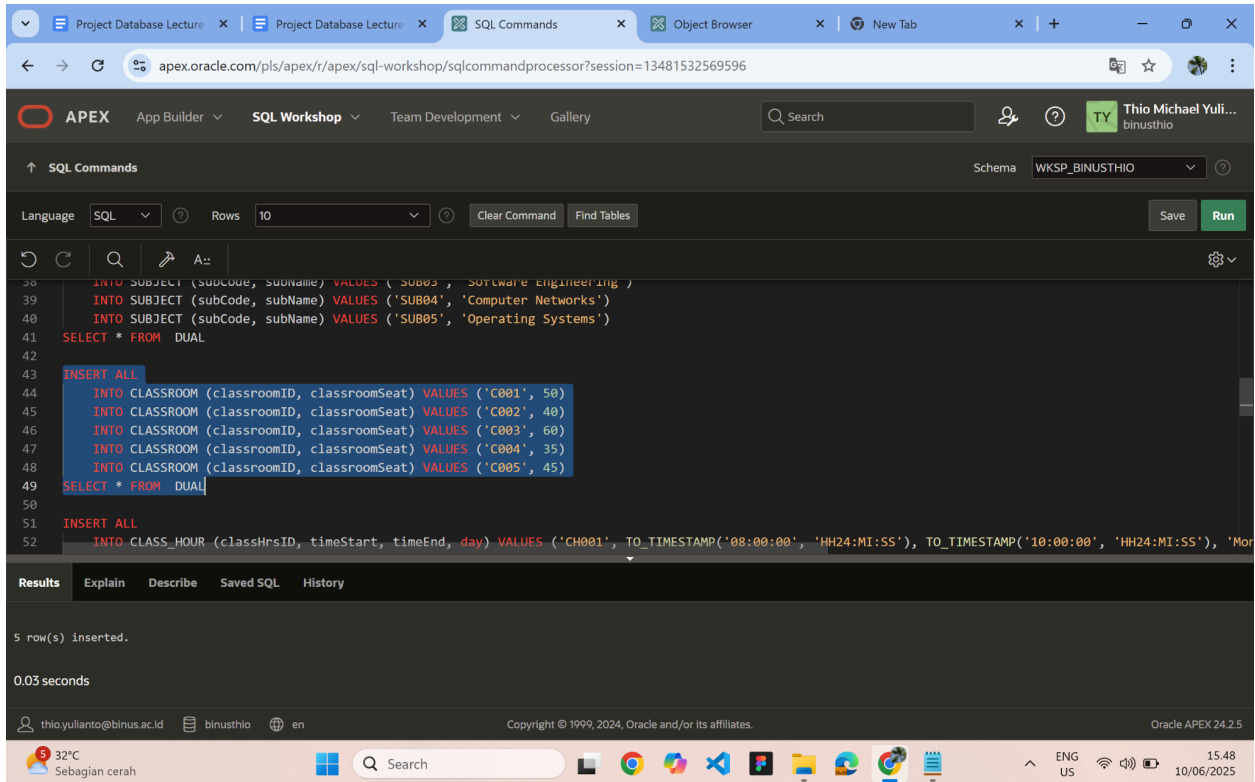
INSERT ALL

```
INTO SUBJECT (subCode, subName) VALUES ('SUB01', 'Database Systems')
```

```
INTO SUBJECT (subCode, subName) VALUES ('SUB02', 'Artificial Intelligence')
```

```
INTO SUBJECT (subCode, subName) VALUES ('SUB03', 'Software Engineering')
INTO SUBJECT (subCode, subName) VALUES ('SUB04', 'Computer Networks')
INTO SUBJECT (subCode, subName) VALUES ('SUB05', 'Operating Systems')
SELECT * FROM DUAL
```

INSERT TABLE CLASSROOM (SAMUEL PRIMA DAMANIK):



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli...' with the schema 'WKSP_BINUSTHIO'. The 'SQL Commands' tab is active, displaying a list of SQL statements. The first three statements are for inserting into the 'SUBJECT' table, and the next five are for inserting into the 'CLASSROOM' table. The 'Run' button is highlighted in green. Below the SQL editor, the 'Results' tab shows the execution outcome: '5 row(s) inserted.' and '0.05 seconds'. The bottom status bar indicates the user is 'thio.yulianto@binus.ac.id', the schema is 'binusthio', and the version is 'Oracle APEX 24.2.5'.

```
38 INTO SUBJECT (subCode, subName) VALUES ('SUB03', 'Software Engineering')
39 INTO SUBJECT (subCode, subName) VALUES ('SUB04', 'Computer Networks')
40 INTO SUBJECT (subCode, subName) VALUES ('SUB05', 'Operating Systems')
41 SELECT * FROM DUAL
42
43 INSERT ALL
44 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C001', 50)
45 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C002', 40)
46 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C003', 60)
47 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C004', 35)
48 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C005', 45)
49 SELECT * FROM DUAL
50
51 INSERT ALL
52 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH001', TO_TIMESTAMP('08:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'), 'Mon
```

Results Explain Describe Saved SQL History

5 row(s) inserted.

0.05 seconds

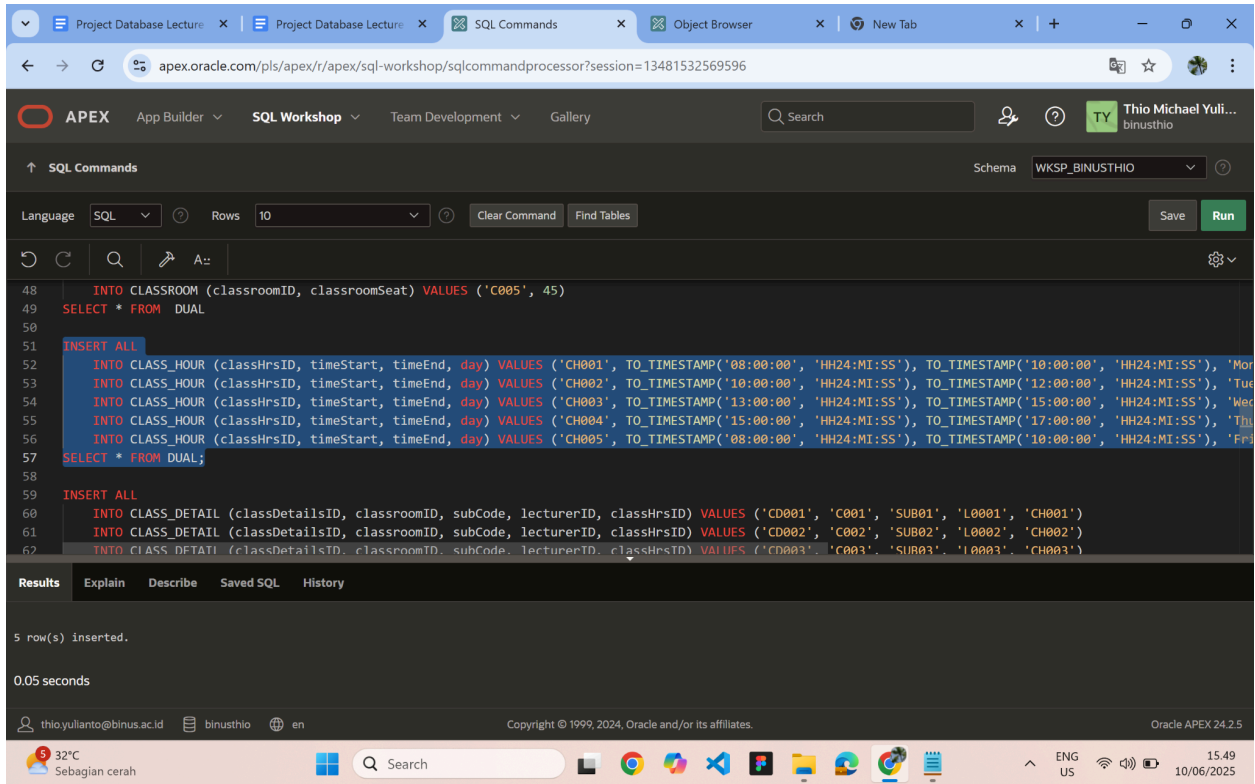
thio.yulianto@binus.ac.id binusthio en Copyright © 1999, 2024, Oracle and/or its affiliates. Oracle APEX 24.2.5

QUERY:

INSERT ALL

```
INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C001', 50)
INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C002', 40)
INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C003', 60)
INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C004', 35)
INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C005', 45)
SELECT * FROM DUAL
```

INSERT TABLE CLASS_HOUR (ANDHIKA HAFIZH ALBANA):



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, displaying a list of SQL commands. The 'Schema' dropdown is set to 'WKSP_BINUSTHIO'. The 'Language' is 'SQL' and 'Rows' is set to '10'. The 'Run' button is highlighted. The SQL commands are as follows:

```
48 INTO CLASSROOM (classroomID, classroomSeat) VALUES ('C005', 45)
49 SELECT * FROM DUAL
50
51 INSERT ALL
52 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH001', TO_TIMESTAMP('08:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'), 'Mon
53 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH002', TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('12:00:00', 'HH24:MI:SS'), 'Tue
54 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH003', TO_TIMESTAMP('13:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('15:00:00', 'HH24:MI:SS'), 'Wed
55 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH004', TO_TIMESTAMP('15:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('17:00:00', 'HH24:MI:SS'), 'Thu
56 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH005', TO_TIMESTAMP('08:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'), 'Fri
57 SELECT * FROM DUAL;
58
59 INSERT ALL
60 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD001', 'C001', 'SUB01', 'L0001', 'CH001')
61 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD002', 'C002', 'SUB02', 'L0002', 'CH002')
62 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD003', 'C003', 'SUB03', 'L0003', 'CH003')
```

The 'Results' tab shows '5 row(s) inserted.' and '0.05 seconds'.

QUERY:

INSERT ALL

INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH001',
TO_TIMESTAMP('08:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'),
'Monday')

INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH002',
TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('12:00:00', 'HH24:MI:SS'),
'Tuesday')

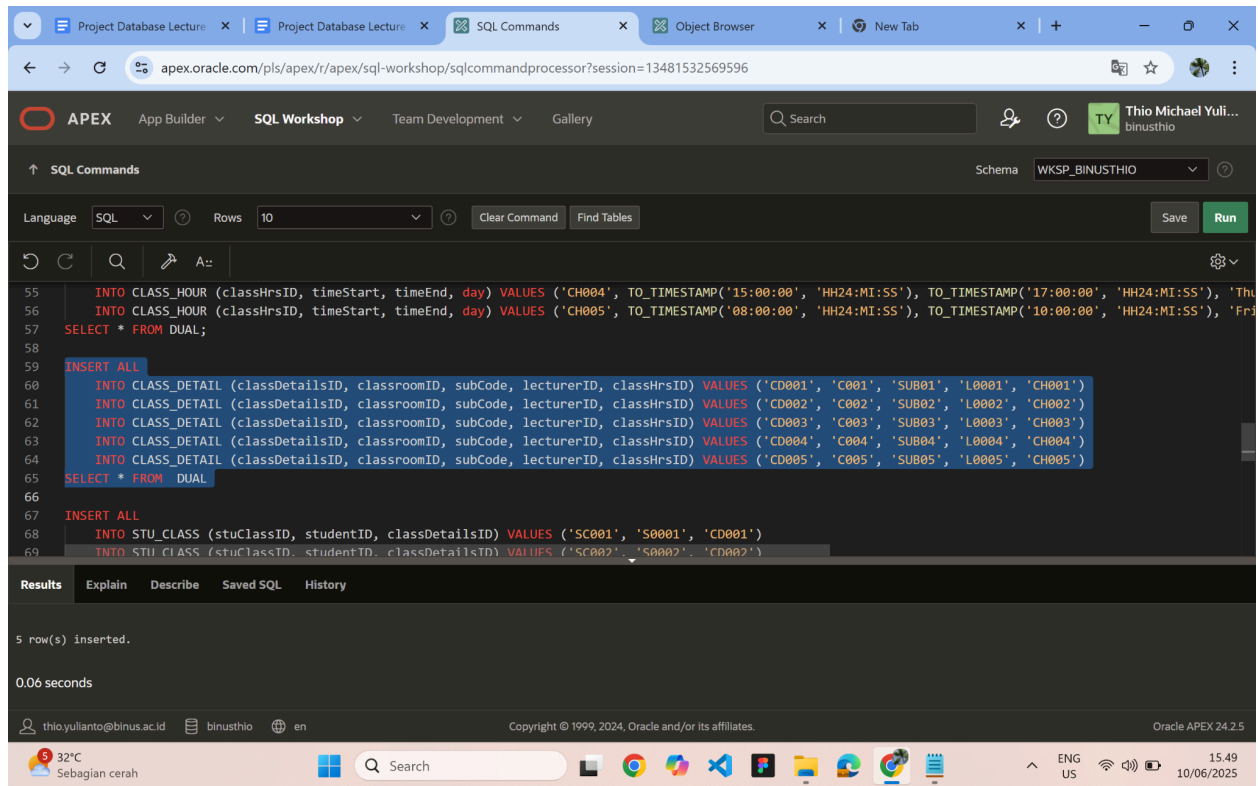
INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH003',
TO_TIMESTAMP('13:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('15:00:00', 'HH24:MI:SS'),
'Wednesday')

INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH004',
TO_TIMESTAMP('15:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('17:00:00', 'HH24:MI:SS'),
'Thursday')

INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH005',
TO_TIMESTAMP('08:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'),
'Friday')

SELECT * FROM DUAL;

INSERT TABLE CLASS_DETAIL (ANDHIKA HAFIZH ALBANA):



```
55 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH004', TO_TIMESTAMP('15:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('17:00:00', 'HH24:MI:SS'), 'Fri
56 INTO CLASS_HOUR (classHrsID, timeStart, timeEnd, day) VALUES ('CH005', TO_TIMESTAMP('08:00:00', 'HH24:MI:SS'), TO_TIMESTAMP('10:00:00', 'HH24:MI:SS'), 'Fri
57 SELECT * FROM DUAL;
58
59 INSERT ALL
60 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD001', 'C001', 'SUB01', 'L0001', 'CH001')
61 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD002', 'C002', 'SUB02', 'L0002', 'CH002')
62 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD003', 'C003', 'SUB03', 'L0003', 'CH003')
63 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD004', 'C004', 'SUB04', 'L0004', 'CH004')
64 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID) VALUES ('CD005', 'C005', 'SUB05', 'L0005', 'CH005')
65 SELECT * FROM DUAL;
66
67 INSERT ALL
68 INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC001', 'S0001', 'CD001')
69 INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC002', 'S0002', 'CD002')
```

Results Explain Describe Saved SQL History

5 row(s) inserted.

0.06 seconds

thio.yulianto@binus.ac.id binusthio en Copyright © 1999, 2024, Oracle and/or its affiliates. Oracle APEX 24.2.5

32°C Sebagian cerah Search ENG US 15.49 10/06/2025

QUERY:

INSERT ALL

 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID)
VALUES ('CD001', 'C001', 'SUB01', 'L0001', 'CH001')

 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID)
VALUES ('CD002', 'C002', 'SUB02', 'L0002', 'CH002')

 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID)
VALUES ('CD003', 'C003', 'SUB03', 'L0003', 'CH003')

 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID)
VALUES ('CD004', 'C004', 'SUB04', 'L0004', 'CH004')

 INTO CLASS_DETAIL (classDetailsID, classroomID, subCode, lecturerID, classHrsID)
VALUES ('CD005', 'C005', 'SUB05', 'L0005', 'CH005')

SELECT * FROM DUAL

INSERT TABLE STU_CLASS (THIO MICHAEL YULIANTO):

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli...' with the schema 'WKSP_BINUSTHIO'. The 'SQL Commands' tab is active, showing a list of commands. The command editor displays the following SQL command:

```
INSERT ALL
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC001', 'S0001', 'CD001')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC002', 'S0002', 'CD002')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC003', 'S0003', 'CD003')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC004', 'S0004', 'CD001')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC005', 'S0005', 'CD002')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC006', 'S0006', 'CD003')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC007', 'S0007', 'CD001')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC008', 'S0008', 'CD002')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC009', 'S0009', 'CD003')
  INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC010', 'S0010', 'CD001')
SELECT * FROM DUAL
```

The command was executed successfully, resulting in 10 rows inserted in 0.06 seconds. The bottom status bar shows the user's email 'thio.yulianto@binus.ac.id', the schema 'binusthio', and the version 'Oracle APEX 24.2.5'.

QUERY:

INSERT ALL

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC001', 'S0001', 'CD001')

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC002', 'S0002', 'CD002')

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC003', 'S0003', 'CD003')

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC004', 'S0004', 'CD001')

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC005', 'S0005', 'CD002')

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC006', 'S0006', 'CD003')

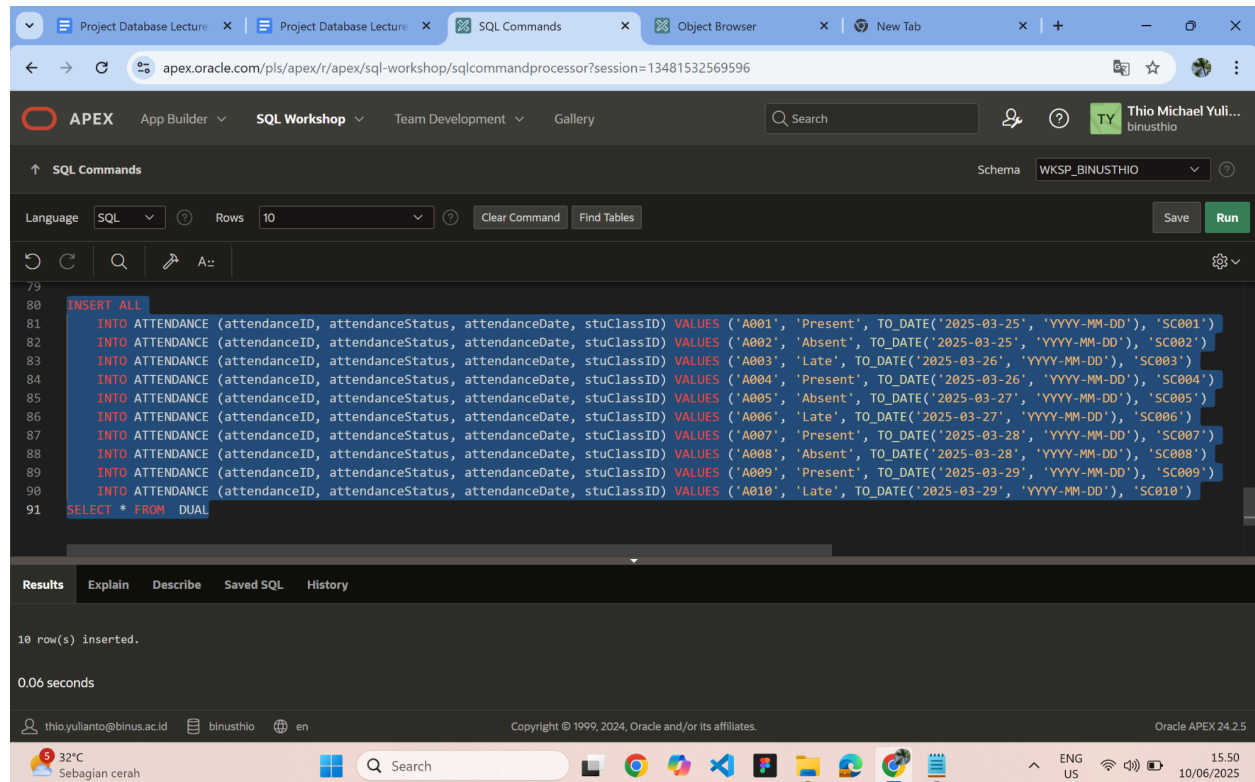
INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC007', 'S0007', 'CD001')

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC008', 'S0008', 'CD002')

INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC009', 'S0009', 'CD003')

```
INTO STU_CLASS (stuClassID, studentID, classDetailsID) VALUES ('SC010', 'S0010',  
'CD001')  
SELECT * FROM DUAL
```

INSERT TABLE ATTENDANCE (THIO MICHAEL YULIANTO):



The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command is as follows:

```
79  
80  
81 INSERT ALL  
82 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A001', 'Present', TO_DATE('2025-03-25', 'YYYY-MM-DD'), 'SC001')  
83 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A002', 'Absent', TO_DATE('2025-03-25', 'YYYY-MM-DD'), 'SC002')  
84 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A003', 'Late', TO_DATE('2025-03-26', 'YYYY-MM-DD'), 'SC003')  
85 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A004', 'Present', TO_DATE('2025-03-26', 'YYYY-MM-DD'), 'SC004')  
86 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A005', 'Absent', TO_DATE('2025-03-27', 'YYYY-MM-DD'), 'SC005')  
87 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A006', 'Late', TO_DATE('2025-03-27', 'YYYY-MM-DD'), 'SC006')  
88 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A007', 'Present', TO_DATE('2025-03-28', 'YYYY-MM-DD'), 'SC007')  
89 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A008', 'Absent', TO_DATE('2025-03-28', 'YYYY-MM-DD'), 'SC008')  
90 INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID) VALUES ('A009', 'Present', TO_DATE('2025-03-29', 'YYYY-MM-DD'), 'SC009')  
91 SELECT * FROM DUAL
```

The results section shows:

```
Results Explain Describe Saved SQL History  
  
10 row(s) inserted.  
  
0.06 seconds
```

QUERY:

INSERT ALL

```
INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)  
VALUES ('A001', 'Present', TO_DATE('2025-03-25', 'YYYY-MM-DD'), 'SC001')
```

```
INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)  
VALUES ('A002', 'Absent', TO_DATE('2025-03-25', 'YYYY-MM-DD'), 'SC002')
```

```
INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)  
VALUES ('A003', 'Late', TO_DATE('2025-03-26', 'YYYY-MM-DD'), 'SC003')
```

```
INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)  
VALUES ('A004', 'Present', TO_DATE('2025-03-26', 'YYYY-MM-DD'), 'SC004')
```

```
INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)  
VALUES ('A005', 'Absent', TO_DATE('2025-03-27', 'YYYY-MM-DD'), 'SC005')
```

```
INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)  
VALUES ('A006', 'Late', TO_DATE('2025-03-27', 'YYYY-MM-DD'), 'SC006')
```

```
INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)  
VALUES ('A007', 'Present', TO_DATE('2025-03-28', 'YYYY-MM-DD'), 'SC007')
```

```

    INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)
VALUES ('A008', 'Absent', TO_DATE('2025-03-28', 'YYYY-MM-DD'), 'SC008')
    INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)
VALUES ('A009', 'Present', TO_DATE('2025-03-29', 'YYYY-MM-DD'), 'SC009')
    INTO ATTENDANCE (attendanceID, attendanceStatus, attendanceDate, stuClassID)
VALUES ('A010', 'Late', TO_DATE('2025-03-29', 'YYYY-MM-DD'), 'SC010')
SELECT * FROM DUAL

```

UPDATE TABLE STUDENT (ADITYA NAUFAL ERLANGGA):

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command being executed is:

```

1 UPDATE STUDENT
2 SET studentTelNo = '081298765432', studentEmail = 'awijaya@gmail.com'
3 WHERE studentID = 'S0001';
4
5 UPDATE FINGERPRINT
6 SET studentID = 'S0009'
7 WHERE studentFPID = 'FP002';
8
9 UPDATE LECTURER
10 SET lecTelNo = '081617633450'
11 WHERE lecturerID = 'L0003';
12
13 UPDATE LECTURER

```

The results section shows:

```

1 row(s) updated.
0.02 seconds

```

The interface also shows the schema WKSP_BINUSTHIO and the user Thio Michael Yuli... binusthio.

QUERY:

UPDATE STUDENT

SET studentTelNo = '081298765432', studentEmail = 'awijaya@gmail.com'

WHERE studentID = 'S0001';

UPDATE TABLE FINGERPRINT (ADITYA NAUFAL ERLANGGA):

The screenshot shows the Oracle APEX SQL Workshop interface. The browser address bar displays the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=101041099120362`. The interface includes a top navigation bar with tabs for SQL Commands, Object Browser, and Project Database Lecture Kel 2. The main area is titled "SQL Commands" and shows a schema of "WKSP_BINUSTHIO". The SQL command editor contains the following code:

```
1 UPDATE STUDENT
2 SET studentTelNo = '081298765432', studentEmail = 'awijaya@gmail.com'
3 WHERE studentID = 'S0001';
4
5 UPDATE FINGERPRINT
6 SET studentID = 'S0009'
7 WHERE studentFPID = 'FP002';
8
9 UPDATE LECTURER
10 SET lecTelNo = '081617633450'
11 WHERE lecturerID = 'L0003';
12
13 UPDATE LECTURER
```

The "Results" tab is selected, showing the output: "1 row(s) updated." and "0.04 seconds". The bottom status bar indicates the user is "thio.yulianto@binus.ac.id" and the system is running Oracle APEX 24.2.5.

QUERY:

UPDATE FINGERPRINT

SET studentID = 'S0009'

WHERE studentFPID = 'FP002';

UPDATE TABLE LECTURER (ARYA RAKA PRATAMA):

The screenshot displays the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli...' with the schema 'WKSP_BINUSTHIO'. The 'SQL Commands' tab is active, showing a list of SQL statements. The command processor is set to 'SQL' and 'Rows' are set to '10'. The command being executed is:

```
1 UPDATE STUDENT
2 SET studentTelNo = '081298765432', studentEmail = 'awijaya@gmail.com'
3 WHERE studentID = 'S0001';
4
5 UPDATE FINGERPRINT
6 SET studentID = 'S0009'
7 WHERE studentFPID = 'FP002';
8
9 UPDATE LECTURER
10 SET lecTelNo = '081617633450'
11 WHERE lecturerID = 'L0003';
12
13 UPDATE LECTURER
```

The 'Results' tab shows the execution outcome: '1 row(s) updated.' and '0.01 seconds'. The bottom status bar indicates the user is 'thio.yulianto@binus.ac.id', the schema is 'binusthio', and the language is 'en'. The copyright notice is 'Copyright © 1999, 2024, Oracle and/or its affiliates.' and the version is 'Oracle APEX 24.2.5'. The system clock shows '18.47' on '10/06/2025'.

QUERY:

UPDATE LECTURER

SET lecTelNo = '081617633450'

WHERE lecturerID = 'L0003';

UPDATE TABLE LECTURER (ARYA RAKA PRATAMA):

The screenshot displays the Oracle APEX SQL Workshop interface. The browser address bar shows the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=101041099120362`. The interface includes a top navigation bar with tabs for SQL Commands, Object Browser, and Project Database Lecture Kel 2. The main area is titled "SQL Commands" and shows a schema of "WKSP_BINUSTHIO". The SQL command editor contains the following code:

```
11 WHERE lecturerID = 'L0003';
12
13 UPDATE LECTURER
14 SET lecPass = 'newlecpass'
15 WHERE lecturerID = 'L0004';
16
17 UPDATE SUBJECT
18 SET subName = 'Cloud Computing'
19 WHERE subCode = 'SUB05';
20
21 UPDATE CLASSROOM
22 SET classroomSeat = 55
23 WHERE classroomID = 'C002';
```

Below the editor, the "Results" tab is active, showing the output: "1 row(s) updated." and "0.00 seconds". The bottom of the interface displays the user "thio.yulianto@binus.ac.id", the schema "binusthio", and the version "Oracle APEX 24.2.5". The Windows taskbar at the bottom shows the date "10/06/2025" and time "18.48".

QUERY:

UPDATE LECTURER

SET lecPass = 'newlecpass'

WHERE lecturerID = 'L0004';

UPDATE TABLE SUBJECT (SAMUEL PRIMA DAMANIK):

The screenshot displays the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli...' with the schema 'WKSP_BINUSTHIO'. The 'SQL Commands' tab is active, showing a list of SQL commands. The command being executed is:

```
15 WHERE lecturerID = 'L0004';  
16  
17 UPDATE SUBJECT  
18 SET subName = 'Cloud Computing'  
19 WHERE subCode = 'SUB05';  
20  
21 UPDATE CLASSROOM  
22 SET classroomSeat = 55  
23 WHERE classroomID = 'C002';  
24  
25 UPDATE CLASS_HOUR  
26 SET timeStart = TO_TIMESTAMP('13:30:00', 'HH24:MI:SS'), timeEnd = TO_TIMESTAMP('15:30:00', 'HH24:MI:SS')  
27 WHERE classHrsID = 'CH003';
```

The 'Results' tab shows the execution outcome: '1 row(s) updated.' and '0.01 seconds'. The bottom status bar indicates the user is 'thio.yulianto@binus.ac.id' with the schema 'binusthio', and the system is running Oracle APEX 24.2.5 on 10/06/2025 at 18.49.

QUERY:

UPDATE SUBJECT

SET subName = 'Cloud Computing'

WHERE subCode = 'SUB05';

UPDATE TABLE CLASSROOM (SAMUEL PRIMA DAMANIK):

The screenshot displays the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli...' with the username 'binusthio'. The schema selected is 'WKSP_BINUSTHIO'. The SQL Commands section shows a query editor with the following SQL code:

```
19 WHERE subCode = 'SUB05';  
20  
21 UPDATE CLASSROOM  
22 SET classroomSeat = 55  
23 WHERE classroomID = 'C002';  
24  
25 UPDATE CLASS_HOUR  
26 SET timeStart = TO_TIMESTAMP('13:30:00', 'HH24:MI:SS'), timeEnd = TO_TIMESTAMP('15:30:00', 'HH24:MI:SS')  
27 WHERE classHrsID = 'CH003';  
28  
29 UPDATE CLASS_DETAIL  
30 SET lecturerID = 'L0005'  
31 WHERE classDetailID = 'CD001';
```

The 'Run' button is highlighted in green. Below the query editor, the 'Results' tab is active, showing the execution results:

1 row(s) updated.
0.02 seconds

The bottom status bar shows the user 'thio.yulianto@binus.ac.id', the schema 'binusthio', and the language 'en'. The copyright notice is 'Copyright © 1999, 2024, Oracle and/or its affiliates.' and the version is 'Oracle APEX 24.2.5'. The system tray at the bottom shows the date '10/06/2025' and time '18.50'.

QUERY:

UPDATE CLASSROOM

SET classroomSeat = 55

WHERE classroomID = 'C002';

UPDATE TABLE CLASS_HOUR (ANDHIKA HAFIZH ALBANA):

The screenshot shows the Oracle APEX SQL Workshop interface. The browser address bar displays the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=101041099120362`. The interface includes a top navigation bar with tabs for SQL Commands, Object Browser, and Project Database Lecture Kel 2. The main area is titled "SQL Commands" and shows a schema of "WKSP_BINUSTHIO". The SQL command editor contains the following code:

```
24
25 UPDATE CLASS_HOUR
26 SET timeStart = TO_TIMESTAMP('13:30:00', 'HH24:MI:SS'), timeEnd = TO_TIMESTAMP('15:30:00', 'HH24:MI:SS')
27 WHERE classHrsID = 'CH003';
28
29 UPDATE CLASS_DETAIL
30 SET lecturerID = 'L0005'
31 WHERE classDetailsID = 'CD001';
32
33 UPDATE STU_CLASS
34 SET classDetailsID = 'CD002'
35 WHERE stuClassID = 'SC001';
36
```

Below the editor, the "Results" tab is active, showing the execution outcome: "1 row(s) updated." and "0.01 seconds". The bottom of the interface displays the user "thio.yulianto@binus.ac.id", the schema "binusthio", and the language "en". The footer includes the copyright notice "Copyright © 1999, 2024, Oracle and/or its affiliates." and the version "Oracle APEX 24.2.5". The system tray at the bottom shows the temperature "29°C Berawan", the search bar, and the date "18.50 10/06/2025".

QUERY:

UPDATE CLASS_HOUR

SET timeStart = TO_TIMESTAMP('13:30:00', 'HH24:MI:SS'), timeEnd =
TO_TIMESTAMP('15:30:00', 'HH24:MI:SS')

WHERE classHrsID = 'CH003';

UPDATE TABLE CLASS_DETAIL (ANDHIKA HAFIZH ALBANA):

The screenshot displays the Oracle APEX SQL Workshop interface. The browser address bar shows the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=101041099120362`. The interface includes a top navigation bar with tabs for SQL Commands, Object Browser, and Project Database Lecture Kel 2. The main area is titled "SQL Commands" and shows a schema of "WKSP_BINUSTHIO". The SQL command editor contains the following code:

```
26 SET timeStart = TO_TIMESTAMP('13:30:00', 'HH24:MI:SS'), timeEnd = TO_TIMESTAMP('15:30:00', 'HH24:MI:SS')
27 WHERE classHrsID = 'CH003';
28
29 UPDATE CLASS_DETAIL
30 SET lecturerID = 'L0005'
31 WHERE classDetailsID = 'CD001';
32
33 UPDATE STU_CLASS
34 SET classDetailsID = 'CD002'
35 WHERE stuClassID = 'SC001';
36
37 UPDATE ATTENDANCE
38 SET attendanceStatus = 'Present'
```

The "Results" tab is selected, showing the output: "1 row(s) updated." and "0.03 seconds". The bottom status bar indicates the user is "thio.yulianto@binus.ac.id" and the system is running Oracle APEX 24.2.5.

QUERY:

UPDATE CLASS_DETAIL

SET lecturerID = 'L0005'

WHERE classDetailsID = 'CD001';

UPDATE TABLE STU_CLASS (THIO MICHAEL YULIANTO):

The screenshot displays the Oracle APEX SQL Workshop interface. The browser address bar shows the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=101041099120362`. The interface includes a top navigation bar with tabs for 'SQL Commands', 'Object Browser', and 'Project Database Lecture Kel 2'. The main area is titled 'SQL Commands' and shows a schema dropdown set to 'WKSP_BINUSTHIO'. The SQL editor contains the following code:

```
30 SET lecturerID = 'L0005';
31 WHERE classDetailsID = 'CD001';
32
33 UPDATE STU_CLASS
34 SET classDetailsID = 'CD002'
35 WHERE stuClassID = 'SC001';
36
37 UPDATE ATTENDANCE
38 SET attendanceStatus = 'Present'
39 WHERE attendanceID = 'A005';
```

Below the editor, the 'Results' tab is active, showing the output: '1 row(s) updated.' and '0.03 seconds'. The bottom of the interface shows the user 'thio.yulianto@binus.ac.id' and the Oracle APEX version '24.2.5'. The Windows taskbar at the bottom indicates the date and time as '10/06/2025' and '18.52'.

QUERY:

```
UPDATE STU_CLASS
SET classDetailsID = 'CD002'
WHERE stuClassID = 'SC001';
```

UPDATE TABLE ATTENDANCE (THIO MICHAEL YULIANTO):

The screenshot shows the Oracle APEX SQL Workshop interface. The browser address bar displays the URL: `apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=101041099120362`. The interface includes a top navigation bar with tabs for SQL Commands, Object Browser, and Project Database Lecture Kel 2. The main area is titled "SQL Commands" and shows a schema dropdown set to "WKSP_BINUSTHIO". The SQL command editor contains the following code:

```
30 SET lecturerID = 'L0005';
31 WHERE classDetailsID = 'CD001';
32
33 UPDATE STU_CLASS
34 SET classDetailsID = 'CD002'
35 WHERE stuClassID = 'SC001';
36
37 UPDATE ATTENDANCE
38 SET attendanceStatus = 'Present'
39 WHERE attendanceID = 'A005';
```

The "Results" tab is selected, showing the output: "1 row(s) updated." and "0.02 seconds". The bottom of the interface displays the user "thio.yulianto@binus.ac.id", the schema "binusthio", and the Oracle APEX version "24.2.5". The Windows taskbar at the bottom shows the date "10/06/2025" and time "18.52".

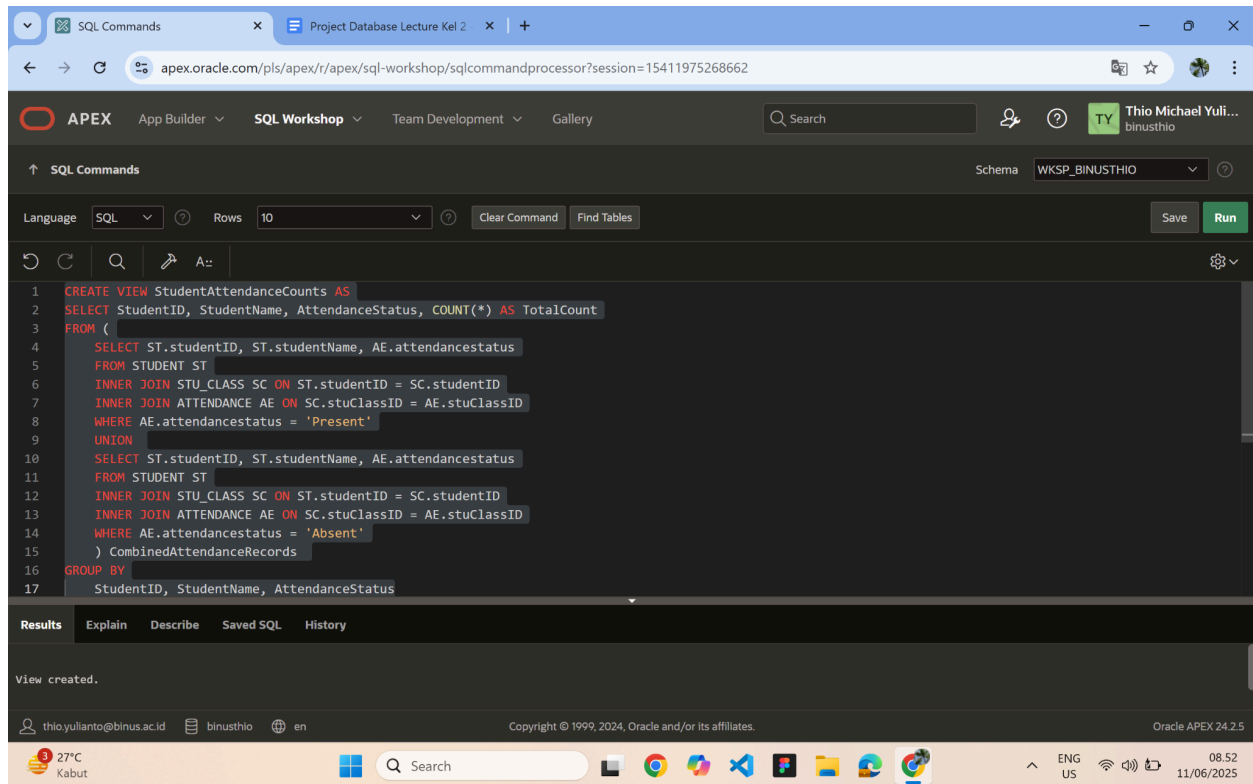
QUERY:

UPDATE ATTENDANCE

SET attendanceStatus = 'Present'

WHERE attendanceID = 'A005';

VIEW



The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The user is logged in as 'Thio Michael Yuli... binusthio'. The 'SQL Commands' tab is active, and the schema is set to 'WKSP_BINUSTHIO'. The SQL editor contains the following code:

```
1 CREATE VIEW StudentAttendanceCounts AS
2 SELECT StudentID, StudentName, AttendanceStatus, COUNT(*) AS TotalCount
3 FROM (
4     SELECT ST.studentID, ST.studentName, AE.attendancestatus
5     FROM STUDENT ST
6     INNER JOIN STU_CLASS SC ON ST.studentID = SC.studentID
7     INNER JOIN ATTENDANCE AE ON SC.stuClassID = AE.stuClassID
8     WHERE AE.attendancestatus = 'Present'
9     UNION
10    SELECT ST.studentID, ST.studentName, AE.attendancestatus
11    FROM STUDENT ST
12    INNER JOIN STU_CLASS SC ON ST.studentID = SC.studentID
13    INNER JOIN ATTENDANCE AE ON SC.stuClassID = AE.stuClassID
14    WHERE AE.attendancestatus = 'Absent'
15    ) CombinedAttendanceRecords
16 GROUP BY
17 StudentID, StudentName, AttendanceStatus
```

The 'Results' tab is selected, showing the message 'View created.' The bottom status bar indicates the user is 'thio.yulianto@binus.ac.id', the schema is 'binusthio', and the version is 'Oracle APEX 24.2.5'. The system clock shows '08:52 11/06/2025'.

QUERY:

```
CREATE VIEW StudentAttendanceCounts AS
SELECT StudentID, StudentName, AttendanceStatus, COUNT(*) AS TotalCount
FROM (
    SELECT ST.studentID, ST.studentName, AE.attendancestatus
    FROM STUDENT ST
    INNER JOIN STU_CLASS SC ON ST.studentID = SC.studentID
    INNER JOIN ATTENDANCE AE ON SC.stuClassID = AE.stuClassID
    WHERE AE.attendancestatus = 'Present'
    UNION
    SELECT ST.studentID, ST.studentName, AE.attendancestatus
    FROM STUDENT ST
    INNER JOIN STU_CLASS SC ON ST.studentID = SC.studentID
    INNER JOIN ATTENDANCE AE ON SC.stuClassID = AE.stuClassID
    WHERE AE.attendancestatus = 'Absent'
    ) CombinedAttendanceRecords
GROUP BY
    StudentID, StudentName, AttendanceStatus
```

HASIL VIEW

SQL Commands

Project Database Lecture Kel 2

apex.oracle.com/pls/apex/r/apex/sql-workshop/sqlcommandprocessor?session=15411975268662

APEX

App Builder

SQL Workshop

Team Development

Gallery

Search

Thio Michael Yuli...
binusthio

SQL Commands

Schema: WKSP_BINUSTHIO

Language: SQL Rows: 10 Clear Command Find Tables Save Run

SELECT *

FROM StudentAttendanceCounts

Results

Explain

Describe

Saved SQL

History

STUDENTID	STUDENTNAME	ATTENDANCESTATUS	TOTALCOUNT
S0007	Gita Anjani	Present	1
S0008	Hadi Susanto	Absent	1
S0005	Eka Putri	Present	1
S0002	Budi Santoso	Absent	1
S0009	Indra Lesmana	Present	1
S0001	Andi Wijaya	Present	1
S0004	Dian Prasetyo	Present	1

7 rows returned in 0.05 seconds Download

thio.yulianto@binus.ac.id binusthio en

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Oracle APEX 24.2.5

27°C Kabut

Search

ENG US 08.53 11/06/2025